Vector Laplacian

$$\nabla^2 \vec{A} = \vec{\nabla}(\vec{\nabla}\cdot\vec{A}) - \vec{\nabla}\times(\vec{\nabla}\times\vec{A})$$

In cartesian coordinates $\nabla^2 \vec{A} = \nabla^2 (A_x \hat{i} + A_y \hat{j} + A_z \hat{k})$

$$= (\nabla^2 A_x)\hat{i} + (\nabla^2 A_y)\hat{j} + (\nabla^2 A_z)\hat{k}.$$

where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$.

However in curvilinear coordinates (cylindrical, spherical) the expressions become more complicated. This is because basis vectors in those systems depend on coordinates as well.

Let $\vec{A} = A_r \hat{r} + A_\theta \hat{\theta} + A_\varphi \hat{\varphi}$ be vector in spherical coordinate system.

Then one finds (after significant work)

Spherical

$$\nabla^2 \vec{A} = \left(\nabla^2 A_r - \frac{2A_r}{r^2} - \frac{2}{r^2 \sin\theta}\frac{\partial(\sin\theta A_\theta)}{\partial\theta} - \frac{2}{r^2\sin\theta}\frac{\partial A_\varphi}{\partial\varphi}\right)\hat{r}$$

$$+ \left(\nabla^2 A_\theta - \frac{A_\theta}{r^2\sin^2\theta} - \frac{2\cos\theta}{r^2\sin^2\theta}\frac{\partial A_\varphi}{\partial\varphi} + \frac{2}{r^2}\frac{\partial A_r}{\partial\theta}\right)\hat{\theta}$$

$$+ \left(\nabla^2 A_\varphi - \frac{A_\varphi}{r^2\sin^2\theta} + \frac{2\cos\theta}{r^2\sin^2\theta}\frac{\partial A_\theta}{\partial\varphi} + \frac{2}{r^2\sin\theta}\frac{\partial A_r}{\partial\varphi}\right)\hat{\varphi}$$

where $\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}$

$$\vec{A} = A_r\hat{r} + A_\varphi\hat{\varphi} + A_z\hat{z}$$

Cylindrical

$$\nabla^2\vec{A} = \left(\nabla^2 A_r - \frac{A_r}{r^2} - \frac{2}{r^2}\frac{\partial A_\varphi}{\partial\varphi}\right)\hat{r} + \left(\nabla^2 A_\varphi - \frac{A_\varphi}{r^2} + \frac{2}{r^2}\frac{\partial A_r}{\partial\varphi}\right)\hat{\varphi}$$

$$+ (\nabla^2 A_z)\hat{z}$$

where $\nabla^2 = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2}{\partial\varphi^2} + \frac{\partial^2}{\partial z^2}$.

Problem 2. A solid spherical ball of radius R charged uniformly with the volume charge density ρ is spinning with angular velocity ω around its axis. Find vector potential $\mathbf{A}(\mathbf{r})$ and magnetic field $\mathbf{B}(\mathbf{r})$ inside and outside the ball. [35 points]

Hint: Look for a solution inside the ball in the form $A_\phi = Cr^3 \sin\theta$.

Useful relations for the spherical system of coordinates:

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2}{\partial \phi^2}$$

$$(\nabla^2 \mathbf{A})_\phi = \nabla^2 A_\phi - \frac{A_\phi}{r^2 \sin^2\theta} + \frac{2}{r^2 \sin\theta}\frac{\partial A_r}{\partial \phi} + \frac{2\cos\theta}{r^2 \sin^2\theta}\frac{\partial A_\theta}{\partial \phi}$$

$$(\nabla \times \mathbf{A})_r = \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(\sin\theta A_\phi) - \frac{1}{r\sin\theta}\frac{\partial A_\theta}{\partial \phi}$$

$$(\nabla \times \mathbf{A})_\theta = \frac{1}{r\sin\theta}\frac{\partial A_r}{\partial \phi} - \frac{1}{r}\frac{\partial}{\partial r}(rA_\phi)$$

$$(\nabla \times \mathbf{A})_\phi = \frac{1}{r}\frac{\partial}{\partial r}(rA_\theta) - \frac{1}{r}\frac{\partial A_r}{\partial \theta}$$

$$\vec{J} = \rho \vec{v} = \rho\, \vec{\omega} \times \vec{r}$$

$$\vec{\omega} = (0, 0, \omega) = \omega \hat{z} = \omega \cos\theta\, \hat{r} - \omega \sin\theta\, \hat{\theta}$$

$$\begin{cases} \hat{r} = \sin\theta \cos\varphi\, \hat{x} + \sin\theta \sin\varphi\, \hat{y} + \cos\theta\, \hat{z} \\ \hat{\theta} = \cos\theta \cos\varphi\, \hat{x} + \cos\theta \sin\varphi\, \hat{y} - \sin\theta\, \hat{z} \\ \hat{\varphi} = -\sin\varphi\, \hat{x} + \cos\varphi\, \hat{y} \end{cases}$$

$$\Rightarrow \cos\theta\, \hat{r} - \sin\theta\, \hat{\theta} = \hat{z}$$

Then $\vec{\omega} \times \vec{r} = (\omega \cos\theta\, \hat{r} - \omega \sin\theta\, \hat{\theta}) \times r\hat{r} = \omega r \sin\theta\, \hat{\varphi}$

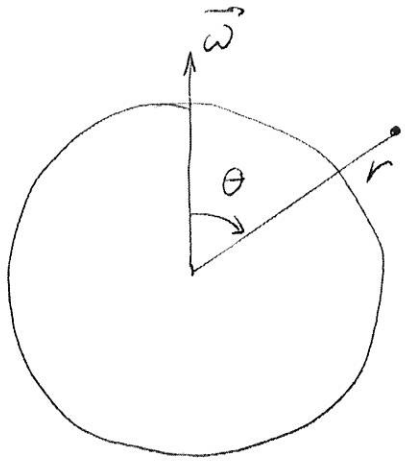
$$\Rightarrow (\nabla^2 \vec{A})_\varphi = -(\mu_0 \vec{J})_\varphi = -\mu_0 \rho \omega r \sin\theta$$

$$(\nabla^2 \vec{A})_\varphi = \nabla^2 A_\varphi - \frac{A_\varphi}{r^2 \sin^2\theta} + \frac{2\cos\theta}{r^2 \sin^2\theta} \cancel{\frac{\partial A_\theta}{\partial \varphi}} + \frac{2}{r^2 \sin\theta} \cancel{\frac{\partial A_r}{\partial \varphi}}$$

$$= \frac{1}{r^2} \frac{\partial}{\partial r}\left(r^2 A_\varphi\right) + \frac{1}{r^2 \sin\theta} \frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial A_\varphi}{\partial \theta}\right) +$$

$$+ \frac{1}{r^2 \sin^2\theta} \cancel{\frac{\partial^2 A_\varphi}{\partial \varphi^2}} - \frac{A_\varphi}{r^2 \sin^2\theta}$$

Problem 2



$$\nabla^2 \vec{A} = -\mu_0 \vec{j} \qquad (1)$$

Current density

$$\vec{j} = \rho \vec{v} = \rho \vec{\omega} \times \vec{r} = (\underset{\uparrow\, r}{0}, \underset{\uparrow\, \theta}{0}, \underset{\uparrow\, \varphi}{\rho \omega r \sin\theta})$$

Since only $j_\varphi \neq 0$ we are going to look for a solution in the form $\vec{A} = (0, 0, A_\varphi)$ as well.

Since there is complete symmetry with respect to rotations around the vertical axis ($\vec{\omega}$) A_φ cannot depend on φ: $A_\varphi = A_\varphi(r, \theta)$

The φ-component of Eq. (1) is

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial A_\varphi}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial A_\varphi}{\partial \theta}\right) - \frac{A_\varphi}{r^2 \sin^2\theta} =$$

$$= -\rho \omega \mu_0 r \sin\theta$$

inside the ball, and the same outside it, except in the latter case the right-hand side is zero (no currents there)

We look for a solution in the form:

$$A_\varphi(r, \theta) = f(r) \sin\theta$$

$$\frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\, f(r) \frac{\partial}{\partial \theta}\sin\theta\right) - \frac{f(r)\sin\theta}{r^2 \sin^2\theta} = -\frac{2}{r^2} f(r) \sin\theta$$

This gives equation for $f(r)$:

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{df}{dr}\right)-\frac{2f}{r^2}=-\rho\omega\mu_0 r\,, \qquad r<R \tag{2}$$

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{df}{dr}\right)-\frac{2f}{r^2}=0\,, \qquad r>R \tag{3}$$

Equation (2) is inhomogeneous, its general solution is the sum of the general solution of the corresponding homogeneous equation (Eq. 3)

$$f_{homo}=ar+\frac{b}{r^2},$$

<u>plus</u> <u>any</u> partial solution of the inhomogeneous equation. The latter we look for in the form cr^λ. Substitution into Eq. (1) yields

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d}{dr}cr^\lambda\right)=c\lambda(\lambda+1)r^{\lambda-2}$$

so that

$$c\lambda(\lambda+1)r^{\lambda-2}-2cr^{\lambda-2}=-\rho\omega\mu_0 r$$

This gives $\lambda=3$ and $c=-\frac{\rho\omega\mu_0}{10}$

The general solution of Eq. (1) is thus

$$f(r)=ar+\frac{b}{r^2}-\frac{\rho\omega\mu_0}{10}r^3 \tag{4}$$

The second term is singular at $r\to 0$, so that $b=0$ inside, $r<R$.

Correspondingly, for $r > R$ only the second term in Eq. (4) should be retained (the other two do not lead to vanishing $\vec{B}$ at $r \to \infty$):

$$f(r) = \begin{cases} ar - \frac{\rho\omega\mu_0}{10} r^3 , & r < R \\ \frac{b}{r^2} , & r > R \end{cases}$$

Since magnetic field is given by derivatives of A_φ, and should be finite everywhere, $f(r)$ and $\frac{df}{dr}$ must be continuous at $r = R$:

$$\begin{cases} aR - \frac{\rho\omega\mu_0 R^3}{10} = \frac{b}{R^2} \\ a - \frac{3\rho\omega\mu_0 R^2}{10} = -\frac{2b}{R^3} \end{cases}$$

This gives

$$a = \frac{\rho\omega\mu_0}{6} R^2 \quad \text{and} \quad b = \frac{\rho\omega\mu_0}{15} R^5$$

We can now write magnetic field $\vec{B} = (B_r, B_\theta, 0)$

$$B_r = \frac{1}{r\sin\theta} \frac{\partial}{\partial\theta} (\sin\theta A_\varphi) = \frac{2}{r} f(r) \cos\theta$$

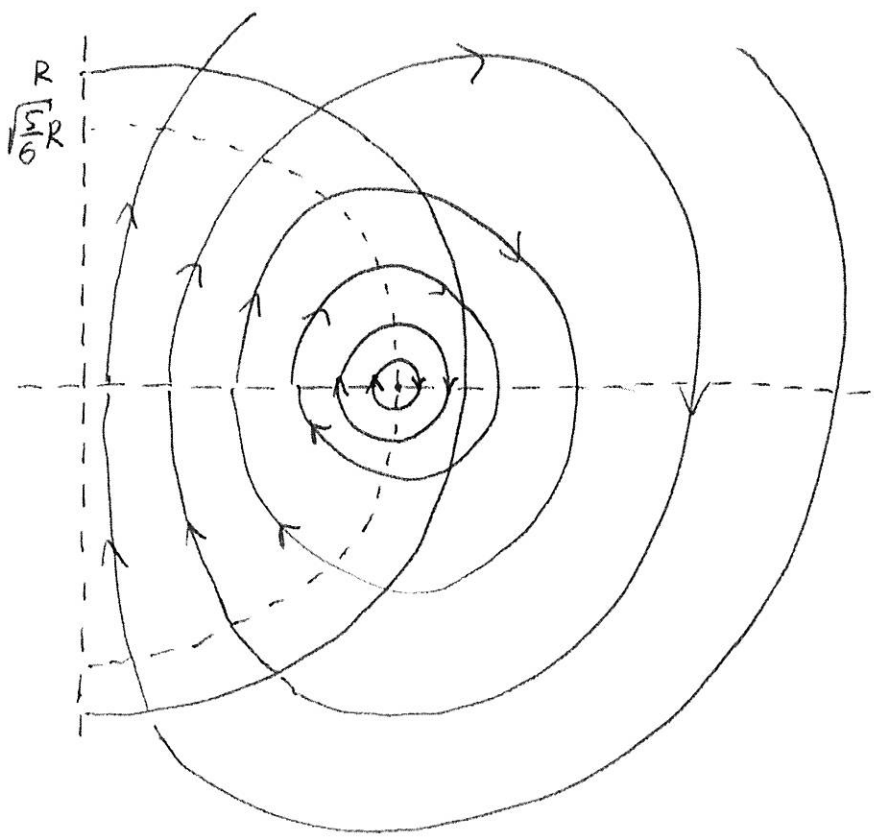
or

$$B_r = \rho\omega\mu_0 R^2 \cos\theta \begin{cases} \frac{1}{3} - \frac{r^2}{5R^2} , & r < R \\ \frac{2}{15} \frac{R^3}{r^3} , & r > R \end{cases}$$

and $\quad B_\theta = -\frac{1}{r}\frac{\partial}{\partial r}(rA_\varphi) = -\sin\theta\left(f' + \frac{f}{r}\right)$

$$B_\theta = \rho\omega\mu_0 R^2 \sin\theta \begin{cases} -\frac{1}{3} + \frac{2}{5}\frac{r^2}{R^2}, & r<R \\ \frac{1}{15}\frac{R^3}{r^3}, & r>R \end{cases}$$

The field outside the ball is the same as the field of a (dipole) magnetic moment

$$\vec{\mu} = \frac{4\pi}{15}\rho\omega R^5 \hat{z}$$

The magnetic field is rather peculiar inside the ball — B_θ changes sign at $r = \sqrt{\frac{5}{6}}\,R$